

LPG Booking System



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1. Introduction:

- The LPG Booking System is a simple, user-friendly digital platform designed to make the process of booking an LPG cylinder easier, faster, and more organized.
- To make the system work effectively, two essential user roles are involved:

1.1. Customer:

- The main user who logs into the system, books LPG cylinder, enters delivery details, makes the payment, and tracks the order status.
- The customer interacts directly with most of the system screens.

1.2. Admin:

1. The administrator who manages customer accounts, oversees bookings, updates cylinder availability, and monitors the overall system activity.
2. The admin ensures that customer requests move smoothly towards delivery.

2. Roles Definition:

The LPG Booking System includes different types of users, each with specific responsibilities and levels of access.

2.1. Role List:

Below is a simple and clear list describing each user role in the system

- **Role Name: Customer**

Description: The primary user who books LPG cylinders for home or commercial use.

Key Permissions:

1. Create an account and log in
2. Book a new LPG cylinder
3. Add or update delivery address
4. Make or confirm payment
5. Track booking and delivery status

- **Role Name: Admin**

Description: The system manager who oversees operations and ensures smooth functioning.

Key Permissions:

- View and manage all customer bookings
 - Update cylinder availability and stock
 - Manage user accounts (approve, block, update information)
 - Monitor payments and verify booking details
-

3. Role Hierarchy:

Here's a simple explanation and diagram that shows how these roles relate:

- **Admin (Top Level)**

The Admin controls the entire system. They manage customers and delivery and ensure everything runs properly. They have the highest authority.

- **Customer (Main User)**

Interacts directly with the system's core features. Customers trigger the booking workflow through their actions.

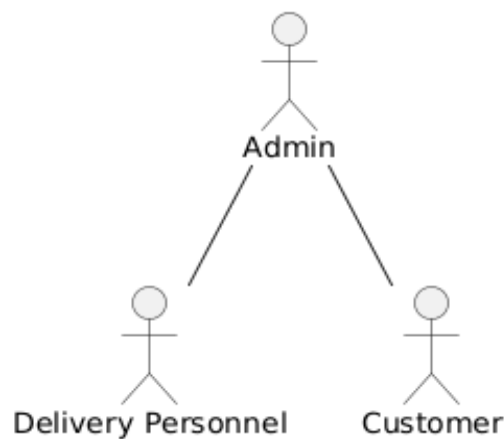


Figure 1: Relationships

3. Mock-ups:

The mock-ups illustrate the key screens designed for each role in the LPG Booking System.

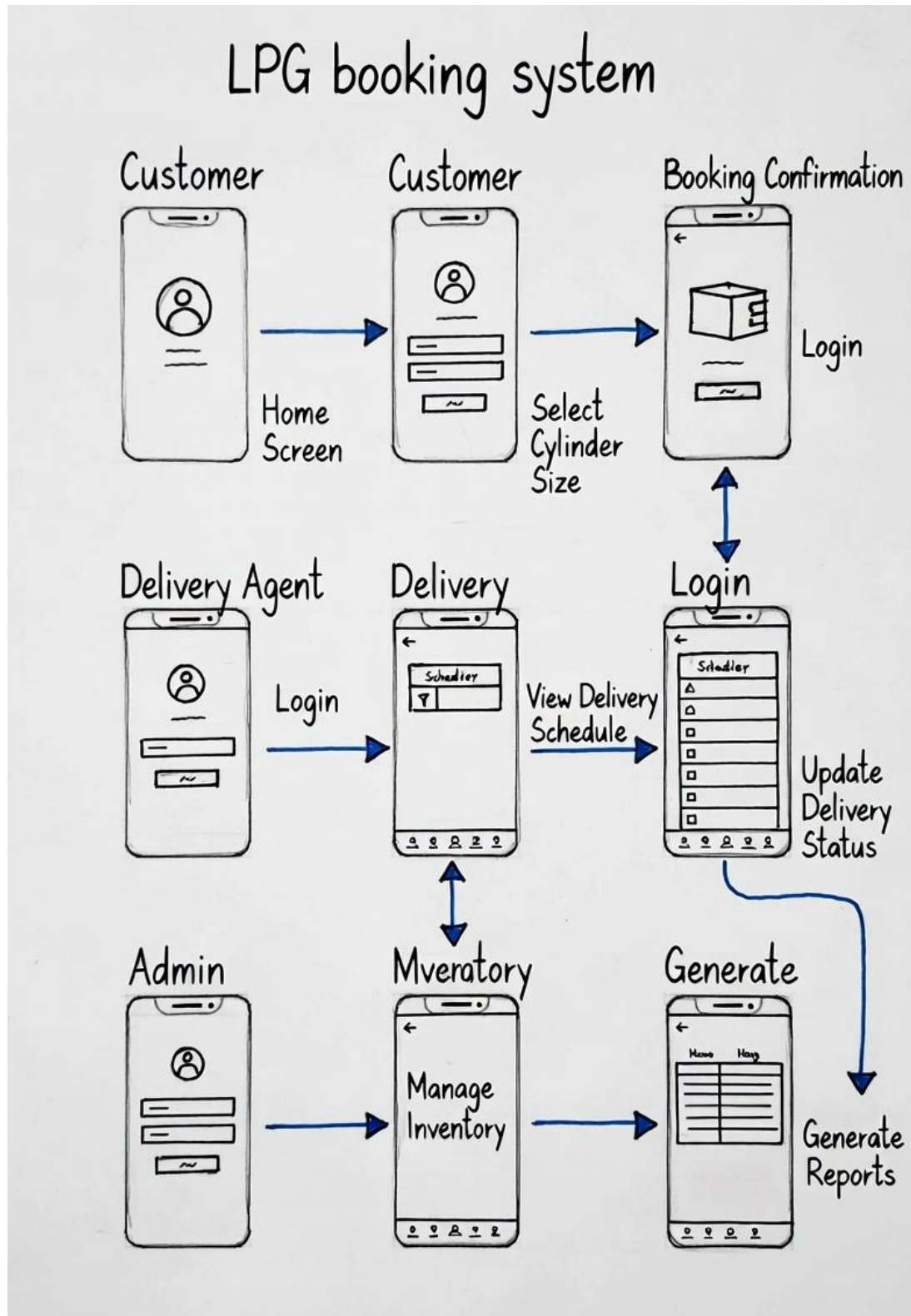


Figure 2: Sketch for each role

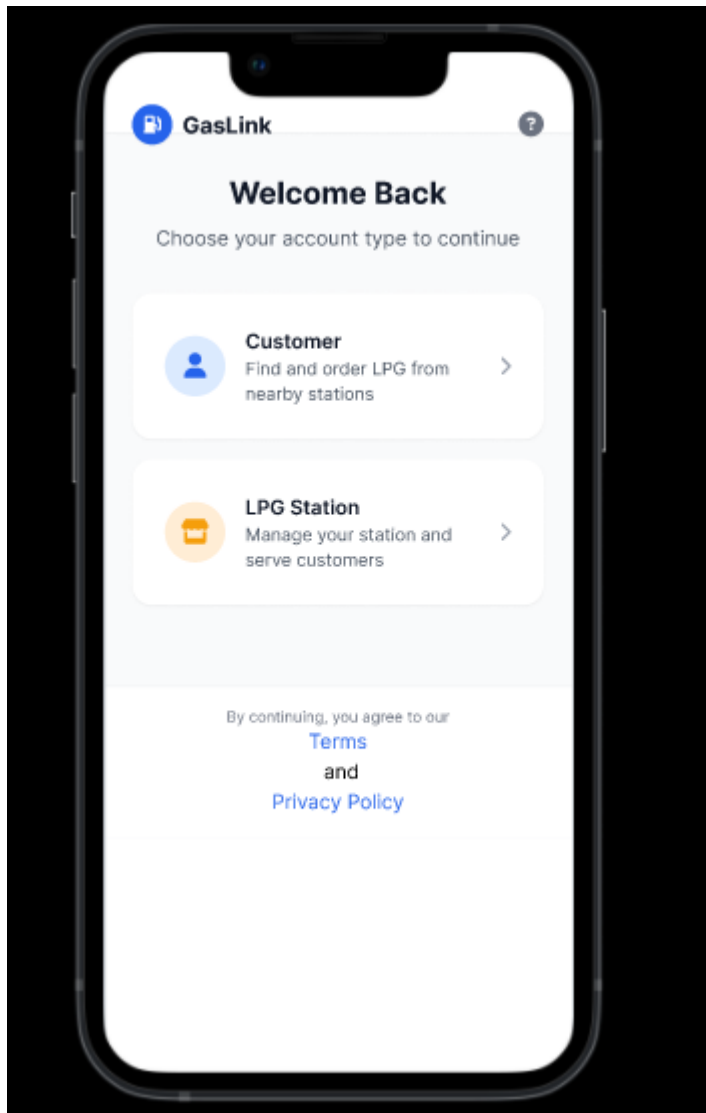
3.1 Customer Mock-ups:

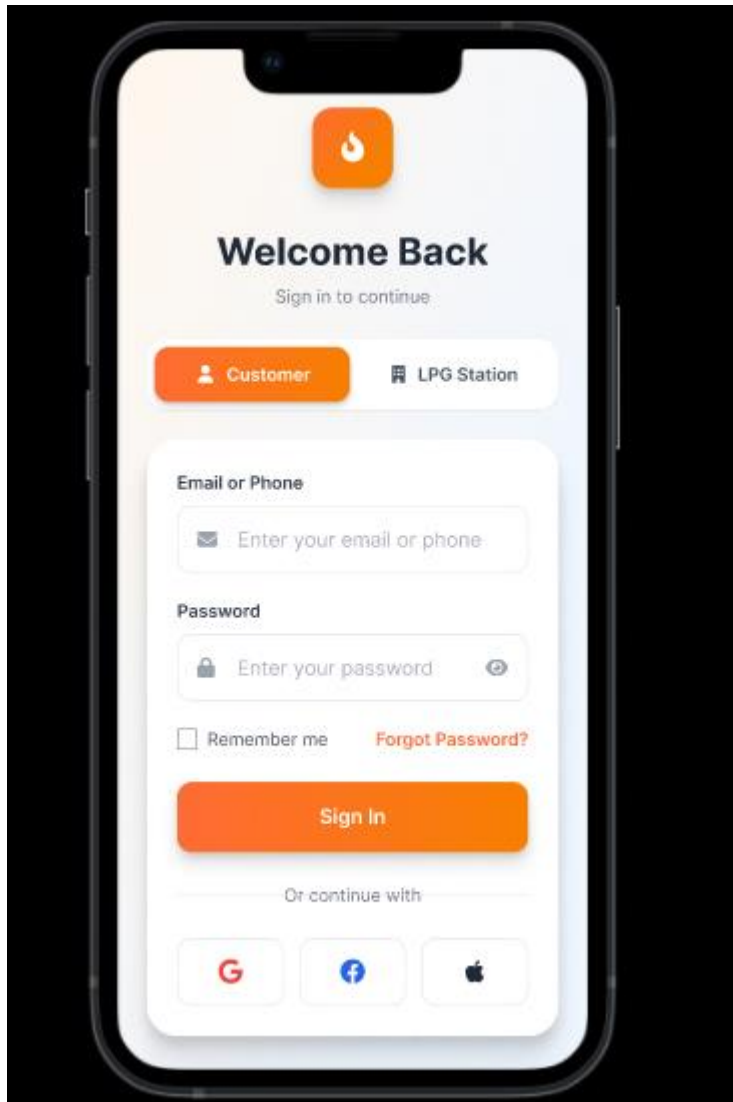
3.1.1 Login & Authentication Process:

Screen 1-3 — Login or Signup

Purpose:

Allows the customer to enter their credentials and access the system.





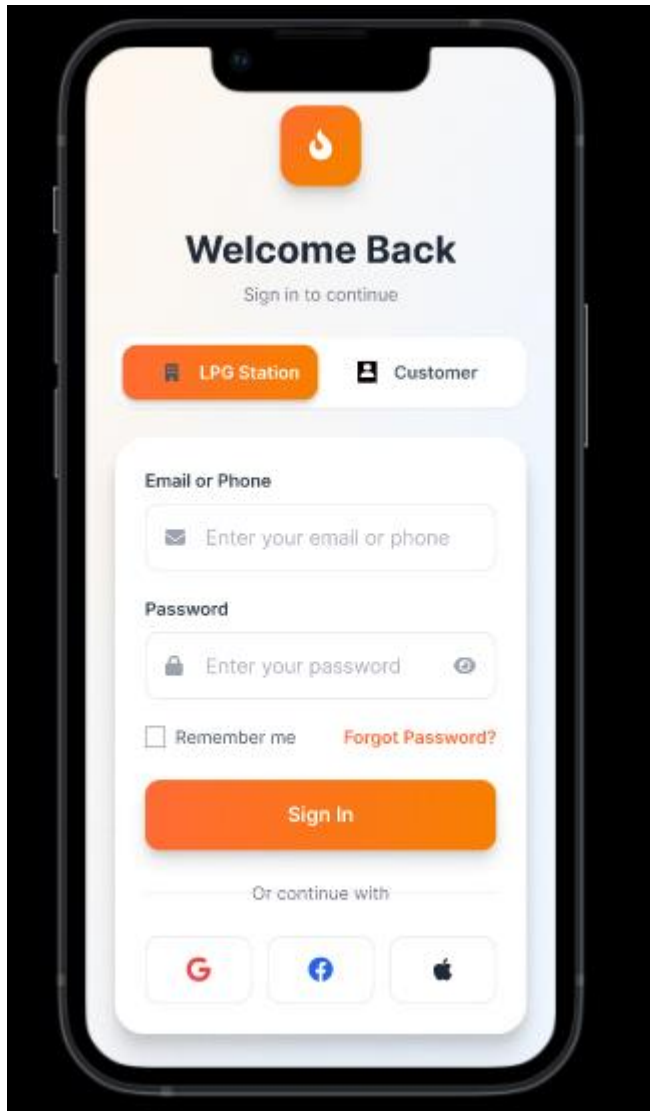


Figure 3: Screen 1

Screen — Signup

Purpose:

Allows the customer to enter their credentials and access the system.

←

Create Account



Welcome!

Create your account to get started with our services and enjoy exclusive benefits.

Full Name



Enter your full name

Email Address



Enter your email

Phone Number



Enter your phone number

Password



Create a password

Must be at least 8 characters long

Confirm Password



Confirm your password

☐ I agree to the

Terms of Service

and

Privacy Policy

☐ I would like to receive marketing emails and updates

Create Account

←

Settings



Sarah Johnson

Customer

>

App Settings



Notifications

>

Legal & Information



Terms & Conditions

>



Privacy Policy

>



About App

v1.2.0

>



Help & Support

>

Logout

Screen —Station Dashboard

Purpose:

Central hub for all admin actions.

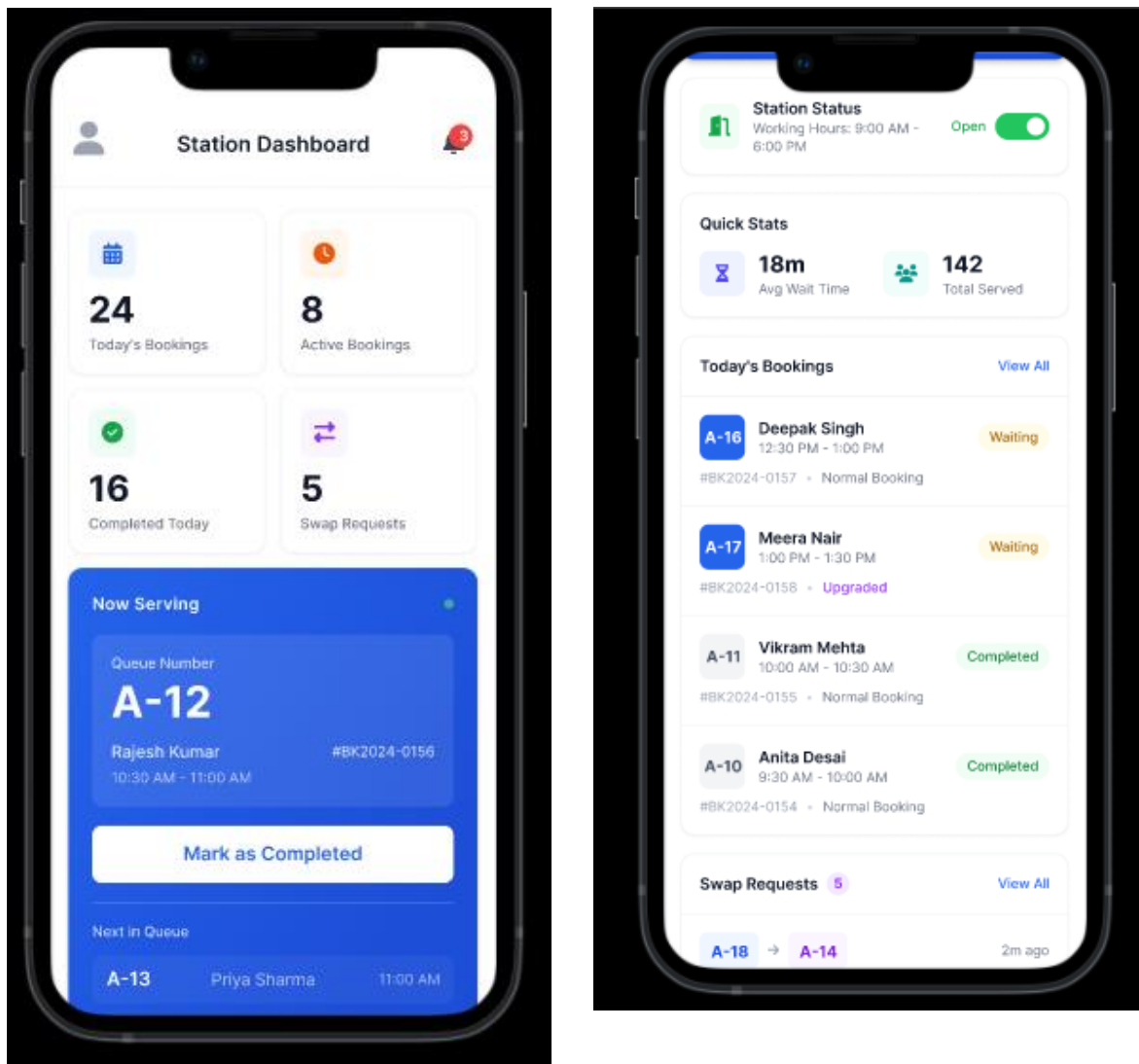


Figure 4: Screen 2 Dashboard

3.1.2 Cylinder Booking Process:

Screen 3 — Bookings

Purpose:

Allows the customer to request a new LPG cylinder.

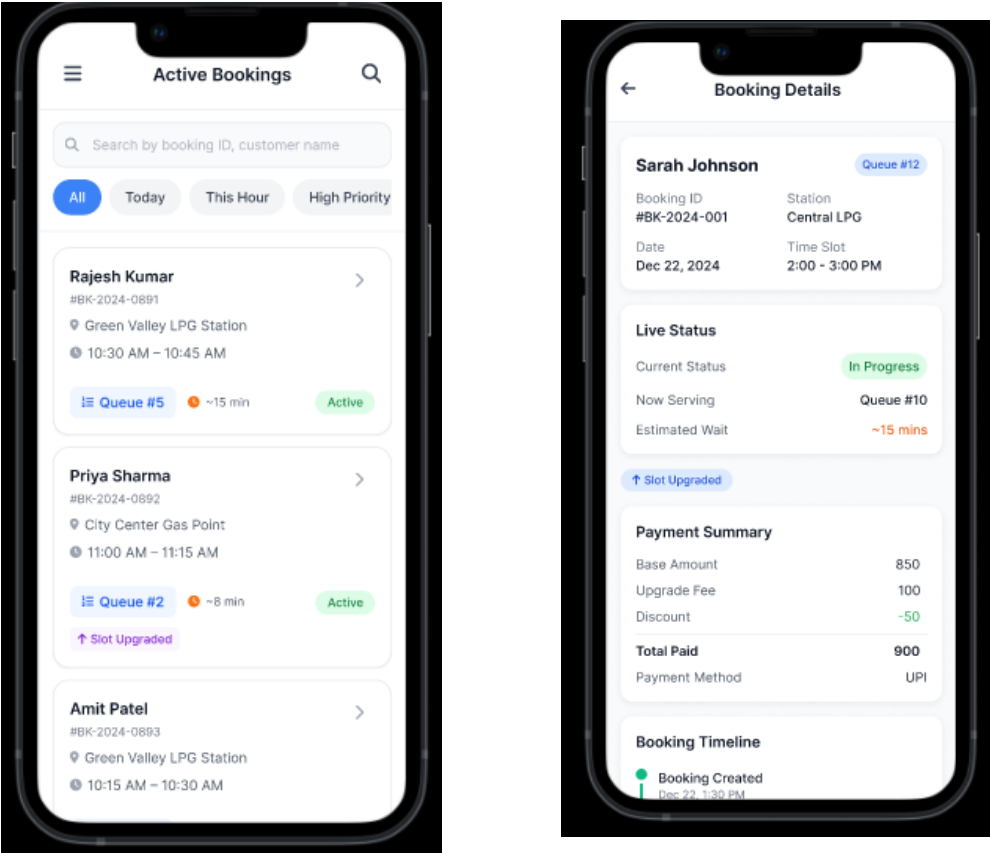


Figure 5:Screen 3 Booking Form

Screen — Payment Screen

Purpose:

Processes the payment securely.

The screenshot displays the 'Payment' screen for 'Bharat Gas Station'. At the top, there is a back arrow and the title 'Payment'. Below this, the station name 'Bharat Gas Station' is shown with a flame icon, followed by the booking ID '#BG2024001'. The selected slot is '10:00 AM - 11:00 AM' and the queue number is '#15'. A 'Price Breakdown' section lists the following items: 'LPG Cylinder Price' at 850.00, 'Slot Upgrade Fee' at 25.00, a 'Discount Applied' of -50.00, and 'Service Charges' at 15.00. The 'Total Amount' is 840.00. Under the 'Payment Method' section, three options are available: 'Credit / Debit Card' (selected with a blue radio button), 'Mobile Wallet' (unselected), and 'Cash on Delivery' (unselected). A green banner at the bottom states 'Your payment is secure and encrypted'. The final 'Pay' button is blue with a white lock icon and the text 'Pay 840.00'.

Price Breakdown	
LPG Cylinder Price	850.00
Slot Upgrade Fee	25.00
Discount Applied	-50.00
Service Charges	15.00
Total Amount	840.00

Payment Method	
☒ Credit / Debit Card Visa, Mastercard, Rupay	☐
☐ Mobile Wallet PhonePe, Paytm, GPay	☐
☐ Cash on Delivery Pay when you receive	☐

Pay 840.00

Figure 6: Screen 4 Payment Screen

Screen — Booking Confirmation Screen

Purpose:

Displays booking ID and success message.

The screenshot displays the 'Create Booking' screen with a back arrow at the top left. The main content is organized into several sections:

- Shell Gas Station**: Includes a location pin icon, '1.8 km away', a 4.2-star rating, and opening hours 'Open: 8:00 AM - 10:00 PM'.
- Select Time Slot**: A grid of six time slots. The '10:15 - 10:30 AM' slot is selected and highlighted in blue. Other slots are marked as 'Available' or 'Full'.
- Queue Information**: Shows a queue number of '#3' and an estimated wait time of '12 min'. A note states: 'Queue number may change based on upgrades or swaps.'
- Booking Summary**: A table summarizing the booking details.

At the bottom, there is a large blue button labeled 'Confirm Booking'.

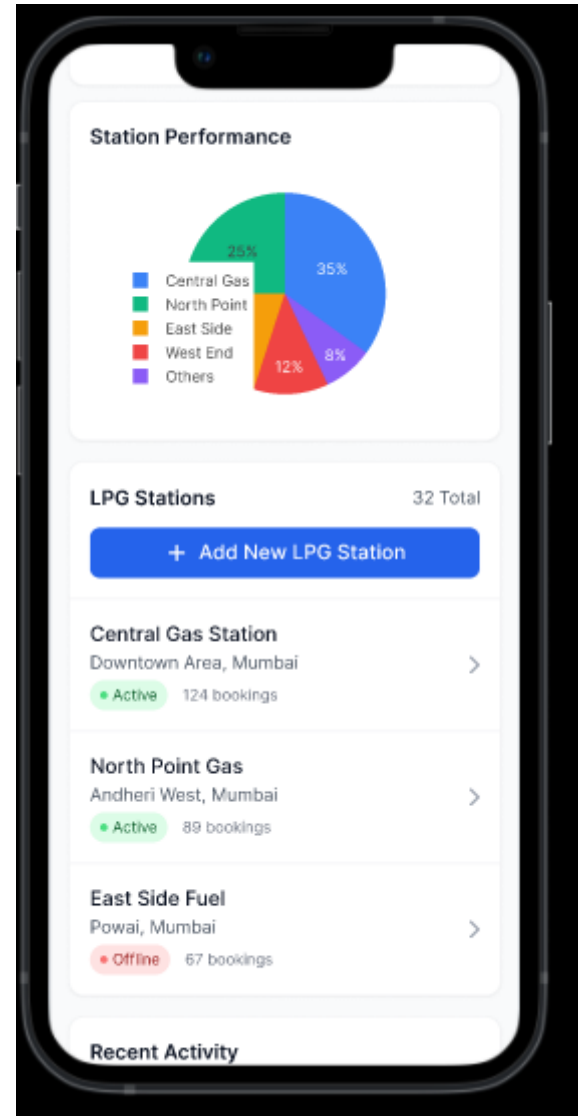
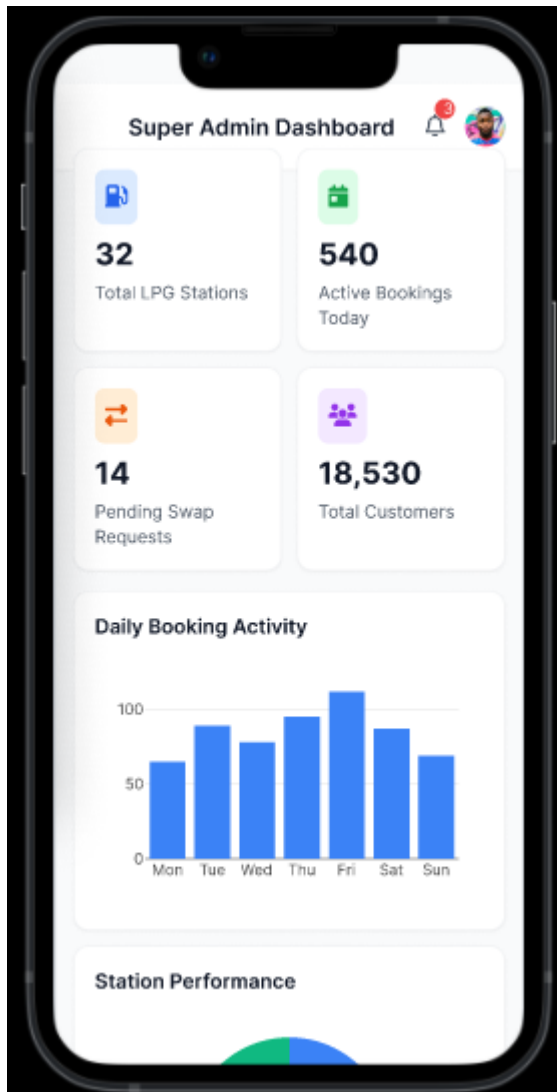
Booking Summary	
Station	Shell Gas Station
Time Slot	10:15 - 10:30 AM
Queue Number	#3
Cylinder Type	14.2 kg Standard
Estimated Total	1500

Figure 7: Screen 5 Booking Confirmation

Screen — Admin Dashboard

Purpose:

Displays admin controls



←

Station Details

Save



Metro Gas Station

ID: STN-001

● Active

Station Information

Station Name

Metro Gas Station

Station Owner

John Smith

Contact Number

+1 234 567 8900

Email Address

metro@gasstation.com

Station Address

123 Main Street, Downtown Area

City / Area

New York

←

Stations



 Search by station name or location

All Stations ▾

Central Gas Station

Active

 Downtown, Mumbai

1234 bookings 

Eastside LPG Hub

Active

 Andheri East, Mumbai

5678 bookings 

North Point Gas

Inactive

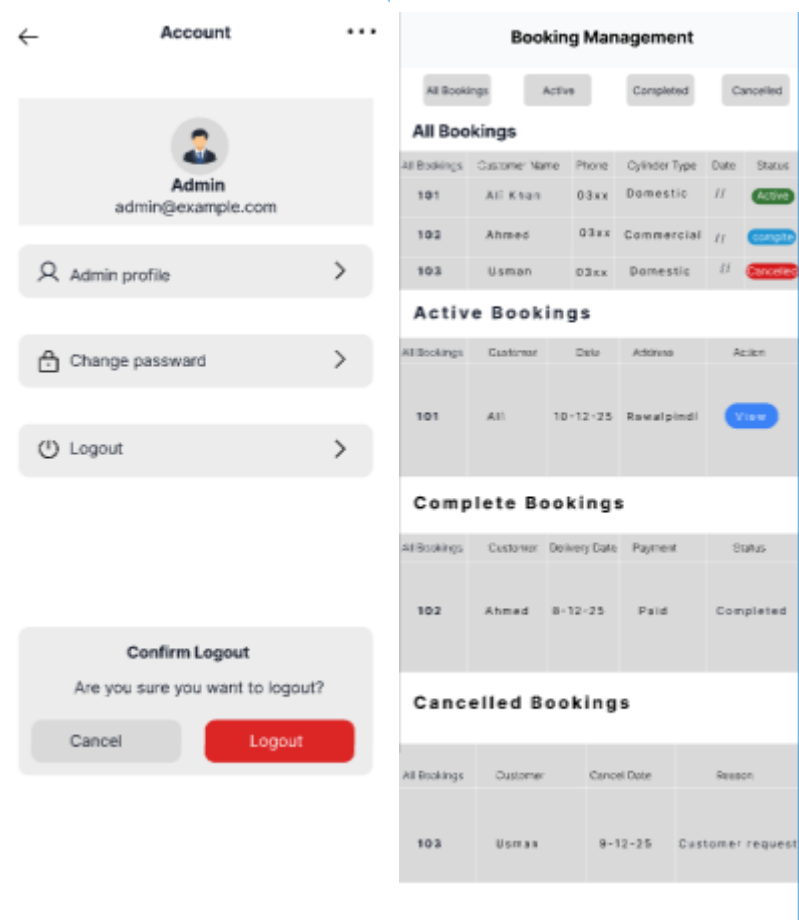
 Borivali West, Mumbai

9012 bookings 

Screen — Admin

Purpose:

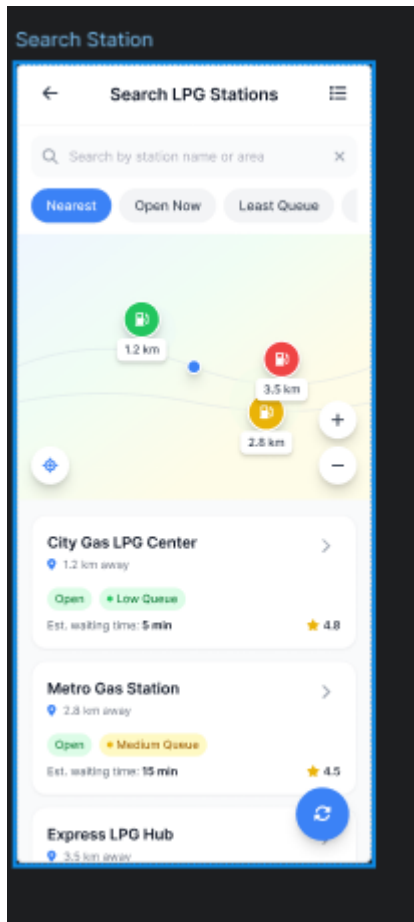
Displays all admin processes.



Screen — LPG Stations

Purpose:

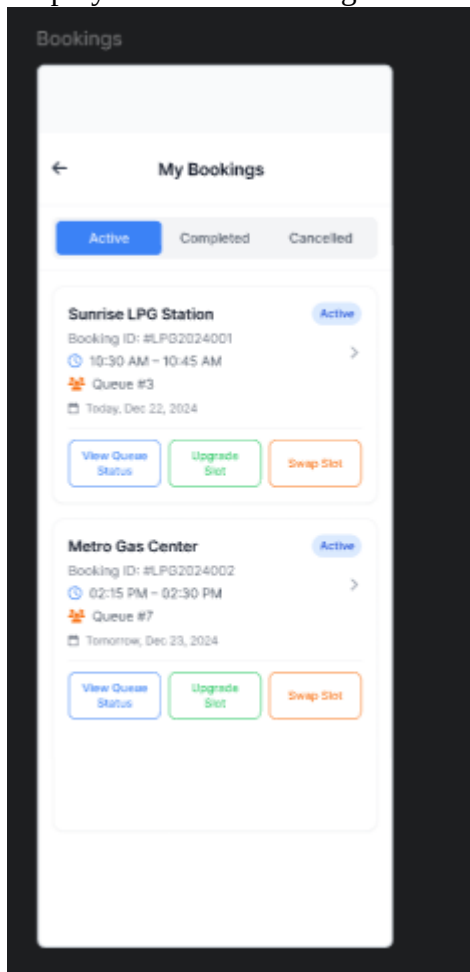
Displays LPG stations list.



Screen — Customer Booking

Purpose:

Displays customer booking.



Screen — Booking Slots

Purpose:

Displays booking slots or swap them.

The screenshot displays a mobile application interface for managing booking slots. At the top, a header bar reads "Swap Slot". Below it, a section titled "Manage Time Slot" includes a back arrow and the text "Current Slot: 10:00 AM — Queue No. 58".

The first section, "Upgrade Slot (Cut Queue)", is accompanied by a clock icon and explains: "Move to an earlier time slot by paying a small fee." It lists three options in light blue boxes:

- 9:30 AM (Queue No. 45) for + Rs 50
- 9:00 AM (Queue No. 32) for + Rs 75
- 8:45 AM (Queue No. 28) for + Rs 100

A blue "Confirm Upgrade" button is positioned below these options.

The second section, "Swap Slot", features a double-headed arrow icon and states: "Exchange your slot with someone who wants a later time." It presents three available swap slots in light green boxes:

- Queue No. 25 ~ 11:00 AM (Get Rs 20 discount) with a selected radio button
- Queue No. 40 ~ 12:00 PM (Get Rs 10 discount) with an unselected radio button
- Queue No. 67 ~ 1:30 PM (Get Rs 15 discount) with an unselected radio button

A green "Request Swap" button is located at the bottom of this section.

A footer note at the bottom left states: "Slot changes are subject to station approval."

Screen — Payment confirmation

Purpose:

Allows the customer to pay.

 **Payment**

**Bharat Gas Station**
Booking ID: #BG2024001

Selected Slot **10:00 AM - 11:00 AM**

Queue Number **#15**

Price Breakdown

LPG Cylinder Price	850.00
Slot Upgrade Fee ⓘ	25.00
Discount Applied	-50.00
Service Charges	15.00
Total Amount	840.00

Payment Method

**Credit / Debit Card**
Visa, Mastercard, Rupay ☒

**Mobile Wallet**
PhonePe, Paytm, GPay ☐

**Cash on Delivery**
Pay when you receive ☐

 Your payment is secure and encrypted

 **Pay**
840.00

4.1 Conclusion:

- The LPG Booking System project demonstrates a simple and effective way to streamline the booking and delivery of LPG cylinders.
 - Through our design, we illustrated how customers can easily log in, book a cylinder, make payments, and track deliveries.
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